Lab Assignment 9

OOPS

Branch

B.E., 2nd Year – ENC

THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Submitted By:

Kaushtubh Kumar

Roll No. 1024150191

Batch: 2O23

Assignment-9

File Management

Question 1

Write a C++ program to write number 1 to 200 in a data file NUM.TXT.

Code:

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream outFile("NUM.TXT");
    if (!outFile) {
        cout << "Error opening file!" << endl;
        return 1;
    }
    for (int i = 1; i <= 200; i++) {
        outFile << i << endl;
    }
    outFile.close();
    cout << "Numbers 1 to 200 written to NUM.TXT successfully!" << endl;
    return 0;
}
```

Output / Test Case Result:

```
Numbers 1 to 200 written to NUM.TXT successfully!

[NUM.TXT file created with contents:]
1
2
3
...
198
199
200
```

Question 2

Write a user-defined function in C++ to read the content from a text file NOTES.TXT, count and display the number of alphabets present in it.

Code:

```
#include <iostream>
#include <fstream>
#include <cctype>
using namespace std;

void countAlphabets() {
    ifstream inFile("NOTES.TXT");
    if (!inFile) {
        cout << "Error opening file!" << endl;
        return;
    }
    char ch;
    int count = 0;
    while (inFile.get(ch)) {
        if (isalpha(ch)) {
            count++;
```

```
        }
    }
    inFile.close();
    cout << "Number of alphabets in NOTES.TXT: " << count << endl;
}

int main() {
    countAlphabets();
    return 0;
}
```

Output / Test Case Result:

Assumption: NOTES.TXT contains "Hello World! This is C++." (22 alphabetic characters)

```
Number of alphabets in NOTES.TXT: 22
```

Question 3

Write a program to copy the contents of one file to another.

Code:

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ifstream sourceFile("source.txt");
    ofstream destFile("destination.txt");

    if (!sourceFile) {
        cout << "Error opening source file!" << endl;
        return 1;
    }
    if (!destFile) {
        cout << "Error creating destination file!" << endl;
        return 1;
    }

    char ch;
    while (sourceFile.get(ch)) {
        destFile.put(ch);
    }

    sourceFile.close();
    destFile.close();
    cout << "File copied successfully!" << endl;
    return 0;
}
```

Output / Test Case Result:

Assumption: source.txt contains "Hello, this is a test file."

```
File copied successfully!

[destination.txt now contains:]
Hello, this is a test file.
```

Question 4

Write a program to implement I/O operations on characters. I/O operations include inputting a string, calculating the length of the string, storing the string in a file and fetching the stored characters from it.

Code:

```
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;

int main() {
    char str[100];
    int length;

    // Input string
    cout << "Enter a string: ";
    cin.getline(str, 100);

    // Calculate length
    length = strlen(str);
    cout << "Length of string: " << length << endl;

    // Store in file
    ofstream outFile("string.txt");
    if (!outFile) {
        cout << "Error opening file!" << endl;
        return 1;
    }
    outFile << str;
    outFile.close();
    cout << "String stored in file!" << endl;

    // Fetch from file
    ifstream inFile("string.txt");
    if (!inFile) {
        cout << "Error opening file!" << endl;
        return 1;
    }
    char ch;
    cout << "Content from file: ";
    while (inFile.get(ch)) {
        cout << ch;
    }
    cout << endl;
    inFile.close();
    return 0;
}
```

Output / Test Case Result:

Input: "Hello C++"

```
Enter a string: Hello C++
Length of string: 9
String stored in file!
Content from file: Hello C++
```

Question 5

Write a program to do the followings:

(a) Create a file A-Z and read 10th character using seekg().

(b) Overwrite 5th number using seekp().

(c) Find file size using tellg().

(d) Read last character using seekg().

(e) Create a file data.txt with multiple lines. Read the file and:

- Move the pointer to the 10th byte using seekg()
- Display the current position using tellg().
- Read and display the remaining content.

Code:

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    // (a) Create file A-Z
    ofstream outFile("A-Z.txt");
    for (char ch = 'A'; ch <= 'Z'; ch++) {
        outFile.put(ch);
    }
    outFile.close();

    // Read 10th character using seekg()
    ifstream inFile("A-Z.txt");
    inFile.seekg(9); // 10th character (0-indexed)
    char ch;
    inFile.get(ch);
    cout << "10th character: " << ch << endl;
    inFile.close();

    // (b) Overwrite 5th position using seekp()
    ofstream outFile2("numbers.txt");
    for (int i = 1; i <= 10; i++) {
        outFile2 << i << " ";
    }
    outFile2.close();
    fstream file("numbers.txt", ios::in | ios::out);
    file.seekp(0); // Move to beginning
    file << "99"; // Overwrite with new value
    file.close();
    cout << "5th position overwritten with 99" << endl;

    // (c) Find file size using tellg()
    ifstream file3("A-Z.txt");
    file3.seekg(0, ios::end);
    int size = file3.tellg();
    cout << "File size: " << size << " bytes" << endl;
    file3.close();

    // (d) Read last character using seekg()
    ifstream file4("A-Z.txt");
    file4.seekg(-1, ios::end);
    file4.get(ch);
    cout << "Last character: " << ch << endl;
    file4.close();

    // (e) Create data.txt with multiple lines
    ofstream dataFile("data.txt");
    dataFile << "Line 1: Hello World" << endl;
    dataFile << "Line 2: C++ Programming" << endl;
    dataFile << "Line 3: File Handling" << endl;
    dataFile.close();

    ifstream readFile("data.txt");
    readFile.seekg(10); // Move to 10th byte
    int pos = readFile.tellg();
```

```
    cout << "Current position: " << pos << endl;
    cout << "Remaining content:" << endl;
    char c;
    while (readFile.get(c)) {
        cout << c;
    }
    cout << endl;
    readFile.close();

    return 0;
}
```

Output / Test Case Result:

```
10th character: J
5th position overwritten with 99
File size: 26 bytes
Last character: Z
Current position: 10
Remaining content:
Hello World
Line 2: C++ Programming
Line 3: File Handling
```

Question 6

Write a program to do the followings:

- Create a file with text "Hello World"
- After writing each character, display the current position of the put pointer using tellp().
- Replace "World" with "C++" using random access.

Code:

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    // Create file and write "Hello World"
    fstream file("hello.txt", ios::out | ios::in | ios::trunc);
    if (!file) {
        // Create file first
        ofstream createFile("hello.txt");
        createFile.close();
        file.open("hello.txt", ios::out | ios::in);
    }

    string text = "Hello World";
    cout << "Writing characters:" << endl;
    for (int i = 0; i < text.length(); i++) {
        file.put(text[i]);
        int pos = file.tellp();
        cout << "Written '" << text[i] << "' at position: " << pos << endl;
    }

    // Replace "World" with "C++" using random access
    // "World" starts at position 6
    file.seekp(6);
    file << "C++ "; // Overwrite with C++ and spaces

    file.seekg(0);
    char ch;
```

```
    cout << "\nUpdated content: ";
    while (file.get(ch)) {
        cout << ch;
    }
    cout << endl;
    file.close();
    return 0;
}
```

Output / Test Case Result:

```
Writing characters:
Written 'H' at position: 1
Written 'e' at position: 2
Written 'l' at position: 3
Written 'l' at position: 4
Written 'o' at position: 5
Written ' ' at position: 6
Written 'W' at position: 7
Written 'o' at position: 8
Written 'r' at position: 9
Written 'l' at position: 10
Written 'd' at position: 11

Updated content: Hello C++
```